COURSE 2 · SECTION 2 · EXERCISE

Which seat ring revision was fitted?

Section 2 · Products, parts, BOM and revisions

Start: Section03_Start.pbix or your Engxtech_S02.pbix; save a copy as Engxtech_S02_Exercise.pbix | **Time:** about 20 min | **Data:** the course data folder

DESIGN QUESTION

Which revision of the P300 valve seat ring was in effect in each build month, does that match what the Builds sheet says was fitted, and how many seat rings did Engxtech fit in total?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

Revisions (loaded in the video) has one row per part revision (PartRev, e.g. P300-B) with its ReleaseDate. Builds has units built per product and month, with the seat ring rev that was fitted. BOM says how many seat rings each product uses. Builds is loaded for real in Section 9; here you load it in your exercise copy only.

New in this exercise (not shown in the video; one line each):

- **FILTER (table, condition)**: returns only the rows of the table where the condition is true (S05 uses it again).
- **ALL (table)**: ignores every filter on the table, so a measure sees all its rows.
- **MAXX (table, expression)**: works out the expression on each row of the table and returns the largest value.
- **LOOKUPVALUE (result column, search column, value, ...)**: returns the result column from the row where each search column equals its value (S08 uses it again).

Your tasks

1. In your exercise copy, load Builds from the course workbook (BuildMonth = Date, UnitsBuilt = Whole number). Revisions and Parts[PartID] 1:* Revisions[PartID] are already there from the video.
2. Model view: add Products[ProductID] 1:* Builds[ProductID] (single direction).
3. In Builds, add a calculated column **RevByDate**: the P300 rev with the latest ReleaseDate on or before BuildMonth.
4. Add a column **RevCheck** that says Match or Different when RevByDate is compared with SeatRingRev. Count each.
5. Write a measure **P300 Rev On Date** for 26 Jan 2026 (ECR-3200 request date); then change the date to 30 Sep 2026. Compare with the video's Current Rev on the Revisions page (AsOfDate slicer).
6. BOM explosion: write **Seat Rings Fitted** = units built x P300 QtyPer of that product, summed over Builds.
7. Explain in one sentence why some build months are Different.

Check your answer

Check	Expected
Rows in Revisions (unique PartRev keys)	**16**
P300 revisions	**3**
P300 rev in effect on 26 Jan 2026	**B**
P300 rev in effect on 30 Sep 2026	**C**
Builds rows where RevByDate = SeatRingRev	**53 of 56**
Different rows	**3 rows, May 2025**
Product units built	**40,962**
Seat rings fitted	**73,503**
Seat rings in one of each product	**5**

Hint. Find the date first (MAXX over a FILTER of Revisions), then look up the Rev for that date with LOOKUPVALUE.

Stuck? The worked solution with the full DAX and M is in Solutions/S02_Solution.pdf.